

Kalamna - AI Customer Support Egyptian service

Smart Customer Support Service for Businesses

Graduation Project Detailed Proposal

Team Members:

Ahmed Elattar, Bassant Hossam, Eyad Hesham, Zeyad Wael,
Shahda Mohamed, Tony Ehab

Supervisor: Dr. Ahmed Yousry

Faculty of Computer Science
Benha National University

Contents

1	Idea Description	1
2	Problem Statement	1
3	Proposed Solution	1
4	Market Research and Competitor Analysis	1
5	MVP and Future Enhancements	3
6	Phases and Roadmap	3
7	Technologies and Team Roles	4

1 Idea Description

Our project is an AI-powered, customizable customer support platform designed to help Egyptian SMEs automate their customer service across channels like WhatsApp, Messenger, and web platforms.

Main Features:

- Understanding of the Arabic dialect of Egypt.
- Voice support (speech-to-text & text-to-speech).
- Real-time emotion detection.
- Multi-intent handling.
- Integrations: WhatsApp, Messenger, Websites, Apps.

2 Problem Statement

Local businesses struggle with:

- providing consistent, 24/7 customer support.
- Repetitive questions,
- missed messages
- slow response times

All this lead to poor customer satisfaction. Existing solutions are often expensive, rigid, or lack authentic local language and dialect support.

3 Proposed Solution

We aim to build a flexible, API-powered Agent/Services system that:

- Uses advanced AI Models & APIs (Gemini, GPT, Whisper) for text, voice, and emotion-aware replies.
- Supports the Egyptian dialect as a primary focus.
- Detects user emotions and adapts responses accordingly.
- Allows businesses to fully customize the bot's tone, responses, and operating hours.

4 Market Research and Competitor Analysis

This chapter provides an overview of key competitors in the AI customer support and chatbot market in Egypt and the wider MENA region. Several companies already offer communication and AI-driven solutions, each with different focuses such as infrastructure, multichannel support, or conversational intelligence.

For more details on chatbot providers in Egypt, see **Clutch.co – Egypt Chatbot Providers**.

Main Competitors:

WideBot

WideBot is a Cairo-based AI startup that provides a no-code chatbot building platform with a strong focus on Arabic and local dialects. The company mainly serves SMEs and enterprises in the MENA region by enabling them to build and deploy chatbots across multiple channels without technical expertise.

Weaknesses:

- Limited advanced AI features such as emotion detection and voice handling.
- Less suited for deep customization.
- Focused more on automation than personalized conversational intelligence.

Focus Areas:

- No-code chatbot creation.
- Support for Arabic and dialects.
- Multi-channel communication (WhatsApp, Messenger, Web).

Main Features:

- Drag-and-drop style dashboard for bot building.
- Automation of FAQs and repetitive queries.
- Basic analytics and reporting.
- Multi-channel deployment support.

Cequens

Cequens is a Middle East-based CPaaS (Communications Platform as a Service) provider that focuses on delivering communication infrastructure and enterprise integration. It mainly provides APIs and dashboards for reliable messaging across multiple channels.

Weaknesses:

- Limited AI specialization.
- Lack of support for Arabic dialects (e.g., Egyptian).
- No advanced emotion detection or tone customization.

Focus Areas:

- Reliable communication channels.
- Enterprise integration.
- Infrastructure over conversational AI.

Main Features:

- SMS, WhatsApp Business, Voice, Email, and Authentication APIs.
- Omnichannel dashboard for managing customer interactions.
- Real-time reporting and analytics.
- Enterprise-grade infrastructure and scalability.

DXwand

DXwand is an Egyptian AI startup founded in 2018 that specializes in conversational AI, knowledge mining, and Arabic dialect awareness. With recent Series A funding, the company is expanding its services across MENA, offering advanced text and voice support for enterprises in telecom, healthcare, e-commerce, and government sectors.

Weaknesses:

- Higher cost and technical complexity.
- Reliance on high-quality training data.
- Privacy and scalability challenges.

Focus Areas:

- Conversational AI for enterprises.
- Arabic dialect and slang support.
- Knowledge mining and document intelligence.

Main Features:

- Text and voice conversational AI that understands multiple Arabic dialects.
- Document Intelligence: analyzing documents to build knowledge bases.
- Offline/on-premises deployment for compliance and security.
- Real-time insights and analytics dashboards.

5 MVP and Future Enhancements

Minimum Viable Product (MVP)

The MVP will focus on delivering a text-only customer service chatbot that supports both Egyptian Arabic and Franco-Arabic. Key elements include:

- Integration with an AI model (NileChat as the primary option, with Gemini Pro 2.5 or Qwen Pro as fallbacks) using prompt engineering and response formatting layers.
- Basic emotion detection with tone adaptation.
- A chatbot interface with a chat window that can be embedded as a widget on any website.
- Company knowledge base integration using Retrieval-Augmented Generation (RAG) supported by PostgreSQL with vector extension.
- Consistent response formatting layer to unify output style across models.

Future Enhancements

Planned enhancements beyond the MVP include:

- Voice support in Egyptian Arabic (speech-to-text and text-to-speech).
- An admin dashboard for businesses to manage their knowledge base & FAQs, customize tone, and view analytics.
- Advanced emotion detection (if not included in the MVP).
- Multi-channel integration with Meta services such as WhatsApp, Facebook, and Instagram.
- Full customization options for businesses, including tone and response management.

6 Phases and Roadmap

The project will be executed across two academic terms. The first term focuses on building and delivering the initial MVP, while the second term extends the system with additional features and demonstrations.

Term 1: MVP Development

1. **System Analysis & Prototype Design** – Gather detailed requirements, refine system analysis, and design a low-fidelity prototype (wireframes, mockups). – Validate requirements with supervisor feedback.
2. **MVP Implementation** – Develop the core backend (FastAPI + PostgreSQL) and frontend chat widget (React). – Integrate basic AI model (NileChat or fallback) with prompt engineering. – Add initial emotion detection, RAG integration, and formatting layers. – Implement company authentication and chat history storage.
3. **Testing the MVP** – Internal testing of core flows (chatbot answering FAQs, emotion detection, RAG retrieval). – Fix bugs and validate performance with simple datasets.
4. **Deployment of MVP** – Deploy on cloud (Railway / Docker). – First demo at the end of Term 1 showing a functional text-based MVP.

Term 2: Extended Development and Demo

1. **Feature Expansion** – Add advanced features: advanced emotion detection, admin dashboard, voice support (Whisper + TTS), and expanded customization options for businesses. – Integrate with WhatsApp, Facebook, and Instagram APIs.
2. **Testing and Refinement** – Conduct user testing with SMEs to evaluate new features. – Measure accuracy, response quality, and user satisfaction.
3. **Deployment of Enhanced Version** – Deploy extended version with added features. – Optimize performance and ensure scalability.

4. **Final Demonstration** – Deliver a comprehensive demo including both the initial MVP and extended features. – Prepare final report and presentation.

Summary Timeline

- **Weeks 1–3 (Term 1):** System analysis, prototype design, infrastructure setup.
- **Weeks 4–8 (Term 1):** MVP development (backend, frontend, AI integration, emotion detection, and RAG).
- **Weeks 9–12 (Term 1):** Internal testing, bug fixing, and initial demo deployment.
- **Term 2 (Months 5–8):** Feature expansion (dashboard, voice, advanced integrations), second round of testing, deployment, and final demo.

7 Technologies and Team Roles

Technologies

- **Backend:** Python, FastAPI
- **Frontend:** React.js (Chatbot interface widget + Dashboard)
- **AI/NLP:** NileChat model, Gemini API, Whisper API, HuggingFace Transformers (via API calls), LangChain, LangGraph, Emotion Detection, RAG, Controller Layer
- **Database:** PostgreSQL with vector extension (for VectorDB functionality)
- **Deployment:** Docker, GitHub CI/CD, Railway Deployment
- **Integrations:** WhatsApp Business API, Facebook API, Instagram API, Web Chat Widget, Admin Dashboard

Team Roles

- **Backend Developers (3):** Responsible for core APIs, databases, pipelines, and also assisting the AI engineer.
- **Frontend Developers (2):** Handle UI/UX, build the chatbot widget and dashboard.
- **AI Engineer (1):** Focused on AI/NLP integration, RAG, emotion detection, and orchestration layers.
- **DevOps:** Tasks (containerization, CI/CD, monitoring) will be handled by the backend team.

Team Information

1. Ahmed Ehab Elattar – Project Manager & Backend Developer

- Email: ahmeddelattarr@outlook.com
- LinkedIn: linkedin.com/in/ahmedelattar-tr
- GitHub: github.com/ahmeddelattarr
- Portfolio: ahmedelattar.tech
- CV: Ahmed Resume PDF
- Specialization: Backend, Python, FastAPI, System Architecture, Project Management, AI/NLP, DevOps, PostgreSQL

2. Bassant Hossam Salah Eldeen – Backend Developer

- Email: bassanthosamm1@outlook.com
- LinkedIn: [linkedin.com/in/bassant-hossam](https://www.linkedin.com/in/bassant-hossam)
- GitHub: github.com/bassanthossamxx
- Portfolio: bassant-portfolio.vercel.app
- CV: Bassant Resume PDF
- Specialization: Backend, Python, FastAPI, PostgreSQL, AI/NLP, DevOps

3. Eyad Hesham – Frontend Developer

- Email: eyadtalha04@gmail.com
- LinkedIn: [linkedin.com/in/eyadhesham](https://www.linkedin.com/in/eyadhesham)
- GitHub: github.com/EyadTalha
- CV: Eyad Resume PDF
- Specialization: UI/UX, React.js, Security Implementation, DevOps

4. Shahda Mohamed Abotalb – Backend Developer

- Email: shahdamohamedabotalb@gmail.com
- LinkedIn: [linkedin.com/in/shahda-mohamed](https://www.linkedin.com/in/shahda-mohamed)
- Portfolio: shahdamohamed.github.io/my-website
- GitHub: github.com/shahdamohamed
- CV: Shahda Resume PDF
- Specialization: Backend, Python, FastAPI, AI/NLP, DevOps

5. Zead Wael – Frontend Developer

- Email: zeyadwaelcs@gmail.com
- LinkedIn: [linkedin.com/in/zead-wael](https://www.linkedin.com/in/zead-wael)
- GitHub: github.com/ZeadWael
- CV: Zead Resume PDF
- Specialization: UI/UX, React.js, Pen-testing, Security Implementation, DevOps

6. Toni Ihab – AI Specialist

- Email: tony0100512@cs.bnu.edu.eg
- LinkedIn: [linkedin.com/in/toni-ihab](https://www.linkedin.com/in/toni-ihab)
- GitHub: github.com/tokihab
- Specialization: AI, ML, NLP, Pen-testing, DevOps